Gujarat University

Department of Computer Science

MCA Semester 3 - Computer Vision

Practical Assignment 4

1. A college wants to monitor its corridor using CCTV. The system should automatically identify when a person starts moving. Develop a Python/OpenCV program using frame differencing to detect motion.

2. A security camera is installed in a parking area where the background remains mostly static. Develop a Python/OpenCV program using background subtraction to detect moving vehicles.

3. A surveillance system needs to detect multiple moving objects in real time. Develop a program using the MOG2 background subtraction technique and display the detected objects.

4. A security company wants to continuously monitor a specific object in a video. Develop an OpenCV program that tracks the selected object as it moves across different frames.

5. A warehouse camera needs to track a particular colored package moving on a conveyor belt. Develop a program using the Mean Shift algorithm to track the package.

6. A sports camera tracks a player whose size and orientation change as the player moves closer to or farther from the camera. Develop a program using the CamShift algorithm to track the player.

7. A sports analyst wants to determine the direction of movement of a player between consecutive video frames. Develop a program using the Lucas-Kanade Optical Flow method.

8. A building security system needs to monitor people entering through an entrance. Develop a program that detects and tracks a person from a video sequence and displays a bounding box around the person.

9. A fitness application records a person performing different exercises. Develop a simple Human Activity Recognition program that identifies activities such as walking, running, and standing.

10. A restricted laboratory is monitored by CCTV. The system should identify unusual human movement and generate an alert. Develop a Computer Vision program that detects abnormal movement from a video sequence.